GWALIOR MUNICIPAL CORPORATION

WATER QUALITY SURVEILLANCE & ENVIRONMENTAL HEALTH CELL - ISO/IEC 17025:201

Central Analytical Testing Laboratory, Maharaj Bada Central Facility, Gwalior (M.P.)

NABL CERTIFIED TC-8491
Statutory BIS IS 10500:2012

BOREHOLE CASING INTEGRITY & GROUNDWATER SURVEY

Hazira deep tube well casing seal inspection and subsurface microbiological assay

REPORT / TRACKING ID:	GWL-BORE-35693	**SAMPLING DATE:**	01 November 2026
MONITORING STATION:	Ward 01 - Hazira Crossing Tube Well #02	**ANALYSIS PROTOCOL:**	Spectrophotometric Batch
ACCREDITED CHEMIST:	Dr. S. K. Gupta (Chief Hydrogeologist, GMC Hydrogeology Cell)	**SECURITY SEAL:**	SSE-KMS AES-256 Encrypted
AWS S3 OBJECT KEY:	`reports/field-sheets/gmc-ward-01/ward01_tubewell_casing_integrity.pdf`		

#	WATER QUALITY PARAMETER	METHOD / INSTRUMENT	BIS IS 10500 LIMIT	RESULT	STATUS
01	**Deep Submersible Depth**	Downhole Ultrasonic Probe	Nominal 120m	**118.5m**	**NOMINAL**
02	**Sanitary Casing Seal Integrity**	Surface Runoff Dye Tracer Method	No Infiltration	**SEALED**	**PASS**
03	**Static Water Table Level**	Electric Sounder Gauge	Baseline 22m	**23.4m**	**NOMINAL**
04	**Fluoride (as F)**	SPADNS Spectrophotometry	Max 1.0 mg/L	**0.92 mg/L**	**PASS**
05	**Total Dissolved Solids (TDS)**	Gravimetric Method	Max 500 mg/L	**480 mg/L**	**PASS**
06	**Nitrate (as NO3)**	Spectrophotometric Screening	Max 45 mg/L	**22.0 mg/L**	**PASS**

STATUTORY VERDICT & FINDINGS:

VERDICT: Subsurface borehole sanitary casing seal intact against surface runoff ingress.

Water quality compliant with Central Ground Water Board (CGWB) potable norms.

EXAMINED & CERTIFIED BY:

Dr. S. K. Gupta (Chief Hydrogeologist, GMC Hydrogeology Cell)

NABL National Accreditation Board Certification #TC-8491

GMC PHED SEAL
VERIFIED & SEALED
SHA-256: e3b0c442...

STATUTORY CLEARANCE AUTHORITY:

GMC Water Works Directorate, Gwalior

Public Health Engineering Department, M.P. Government